

Raven Avatar Design Brief

IOM — Interactive Object Media · CRM Screen Recorder (PiP)

Assets for realtime WebGL · Head required · Body optional (Phase 2)

1. Goal

Deliver realtime **3D raven assets** for a browser screen-recorder picture-in-picture:

1. **Head (required)** — speaking beak puppet
2. **Body (optional / Phase 2)** — full raven for larger PiP or idle presence

Speech is driven in-app from microphone / face tracking (beak open–close). Do **not** bake talking animations into the file.

Use case: WebGL (Chrome/Edge), ~30 fps · Small on-screen size (head often ~150–250 px) · Silhouette and beak must read clearly when small.

AI / Meshy generation: Prompt and reference images must show a **clearly open beak** (visible gap between upper and lower beak). A closed beak usually fuses into one solid mesh and is hard to animate. Final delivered GLB may rest closed, but the open shape must exist as bones or a `jawOpen / beakOpen` morph (0 = closed, 1 = open).

2. Deliverables

Required — Head

- `raven-head.glb`
- Short note: bone/morph names + beak open axis/direction
- Optional turntable preview (MP4/GIF)

Optional — Body (Phase 2)

- `raven-body.glb` or `raven-full.glb` (head+body)
- Note: hierarchy, root, wing/leg bones if any
- Optional idle pose turntable

Prefer **two files** (head-only + full) so the app can use head for tiny PiP and body when space allows.

3. Part A — Raven head (required)

Include

- Head + neck only (cut at mid-neck / collar)
- Clear beak, eyes, crown — **beak must be separable for speaking**
- For AI sculpt: generate with **beak open** (gap visible); then rig so closed = rest
- PBR (baseColor + optional roughness/metal/normal)

Avoid

- Full body in this file
- Heavy hair cards / special shaders
- Baked speech animation
- Closed fused beak with no gap / no separate upper-lower parts
- Huge textures (prefer total $\leq 2\text{--}4$ MB; 1K–2K maps)

Rigging — pick A or B (A preferred)

A — Bones (preferred)	B — Morph targets
Head BeakUpper and/or BeakLower (or single Beak) Rest pose = beak closed Document open rotation axis + direction	Exact name: jawOpen or beakOpen 0 = closed, 1 = fully open Optional: eyeBlinkLeft , eyeBlinkRight

Naming / scale / origin

- Root e.g. RavenHead · no spaces in names
- Freeze transforms · facing **+Z toward camera** (or document)
- Origin at **neck base** or **head center** — state which
- Rough head height **0.2–0.4 m** (1 unit $\approx$ 1 m OK)

Performance

- **3k–15k triangles** (hard cap $\sim 30k$)
- 1–2 materials if possible
- File ideally **under ~5 MB** (soft max ~ 15 MB)

4. Part B — Raven body (optional / Phase 2)

Full raven for PiP / mascot. Speech still uses the **beak**; body may get light idle motion in code later — prefer clean bind pose. Optional short idle clip is OK.

Include

- Full body: torso, wings (folded preferred), legs, tail, head + beak
- Same art style and scale as the head asset
- PBR · readable silhouette at small size

Prefer for PiP

- **Folded wings** (compact bounds)
- Neutral standing or perched pose
- Tight bounding box

Avoid

- Spread-wing flight as default pose
- Ultra-dense feathers / 100k+ tris
- Physics / cloth · many draw calls

Rigging (body)

Minimum:

- Root: `RavenRoot`
- `Spine` / `Chest` / `Neck` / `Head`
- Beak same as head brief
- Optional: `WingLeft` , `WingRight` , `LegLeft` , `LegRight` , `Tail`

Nice to have: looping clip named `Idle` (2–4 s, subtle breathe/weight shift).

Performance (body)

- **15k–40k tris** (cap ~60k)
- Textures total ideally $\leq$ 6–8 MB
- File ideally **under ~10 MB** (soft max ~20 MB)
- $\leq$ 3–4 materials

Orientation

- Same as head: **+Z face camera**, Y up
- Origin at **ground between feet** or **perch contact** — document which

5. Shared export checklist

- ☐ Export glTF 2.0 Binary (`.glb`)
- ☐ Include skins / morphs used
- ☐ Apply modifiers; freeze scale
- ☐ No unused lights/cameras
- ☐ Opens in <https://gltf-viewer.donmccurdy.com/>
- ☐ Beak open/close works (bone or morph)
- ☐ Looks clear when scaled to ~200 px wide

6. Acceptance tests

Head	Body
Loads alone Beak opens/closes Readable in a small circle crop	Loads alone Compact folded pose fits tall PiP Beak still controllable No extreme origin offset

7. Out of scope for designer

- Mic / MediaPipe wiring · recorder UI
- Phoneme lip-sync (only open/close intensity)
- Flight / walk cycles (unless explicitly requested later)

8. Paste-ready prompt for designer / Meshy

Design and deliver realtime WebGL raven assets for a screen-recorder PiP.

Required: raven-head.glb — head + neck only, PBR, 3k–15k tris, under ~5 MB, facing camera.

CRITICAL — speaking beak: Generate / sculpt with the beak CLEARLY OPEN (visible gap between upper and lower beak). Deliver controllable open–close via bones (BeakUpper/BeakLower) or morph jawOpen/beakOpen (0=closed, 1=fully open). Do not fuse the beak into one solid closed shape. No baked speech animation.

Optional Phase 2: raven-full.glb — full body, folded wings, standing/perched, same style/scale as head, 15k–40k tris, under ~10–20 MB. Same beak controls. Optional subtle looping Idle clip. Root at ground/perch. Y-up, +Z toward camera.

Deliver GLBs + a short note listing bone/morph names and beak open axis. Test in Don McCurdy's glTF viewer before handoff.

Meshy Text to 3D prompt:

Stylized common raven head and neck only, cut at mid-neck, no body no wings no legs. Side three-quarter view. Beak CLEARLY OPEN with a visible gap between upper and lower beak, large readable beak for speaking puppet. Sharp silhouette, dark black feathers with subtle blue sheen, simple game-ready look, clean topology, solid color not sketch, white background, no text no watermark

9. References (client to attach)

- Brand raven images / existing site raven
- Style preference: realistic vs stylized mascot
- Delivery order: head first, body second (recommended)